

Pharmacy Stock Redistribution & Recommendation System

Product Name: PharmaShift – Expiry-Aware Pharmacy Network Optimizer

Project Title	Pharmacy Stock Redistribution & Recommendation System
Product	PharmaShift – Expiry-Aware Pharmacy Network Optimizer
Review Milestone	Project Review #2 (Minimum 70% Completion)
Assessed Completion	78.5% (Exceeds 70% threshold)
Previous Review	Review #1 – 35% Milestone
Review #1 Score	34.3 / 35 marks (98% criteria met)
Automated Tests	30 / 30 Passed (100% pass rate)
Production Build	Vite Production Bundle (0 errors)
Report Date	September 28, 2026
Submission Deadline	October 5, 2026

Student & Department Placeholders:

- Student Name(s): [Student Name(s) Placeholder]
- Register / Roll Number(s): [Register / Roll Number(s) Placeholder]
- Project Supervisor / Guide: [Project Supervisor Name Placeholder]
- Department of Computer Science & Engineering / Information Technology

2. Executive Summary

PharmaShift is an AI-assisted pharmacy stock redistribution and recommendation system designed to reduce medicine expiry, excess inventory, stock imbalances, and avoidable financial waste across multi-branch pharmacy networks. The system analyzes inventory batches, expiry dates, demand velocity, safety buffers, transit distances, and storage capacities to generate explainable, auditable redistribution recommendations.

Following Review #1 (which scored 34.3 / 35 marks, 98% criteria met), the project has progressed to **78.5% overall completion**, exceeding the Review #2 minimum 70% threshold. The completed core pipeline is verified by 30 automated backend tests (100% passing), a zero-error production Vite build (2,353 modules), and empirical simulation validation across 30 scenario runs.

3. Problem Statement & Methodology

Retail pharmacy networks suffer massive financial losses when near-expiry medicines sit unconsumed in low-demand branches while high-volume branches experience concurrent stockouts. Without intelligent redistribution, usable stock expires unused. PharmaShift solves this through: *Demand Prediction + Expiry Awareness + Inventory Analysis + Safety Constraints + Destination Ranking + Human Approval*.

4. Project Objectives

- 1. Continuously monitor inventory batches across 18 pharmacy branches.
- 2. Detect expiry risks relative to a deterministic reference date (2026-08-14).
- 3. Compute local surplus while reserving a mandatory 3-day safety stock buffer.
- 4. Predict destination dispensing demand using a Random Forest ML model.
- 5. Evaluate Haversine transit distances and enforce a 3-day post-transit shelf life buffer.
- 6. Dynamically rank destination pharmacies using ML predictions and distance penalties.
- 7. Size feasible, capacity-constrained transfer quantities.
- 8. Provide explainable recommendations with a normalized 0-100 Recommendation Score.
- 9. Enforce human clinical governance (Approve, Reject with reasons, Quantity Override).
- 10. Maintain an immutable compliance audit trail capturing state transitions.
- 11. Empirically benchmark savings against a FIFO baseline across 30 simulation cycles.
- 12. Provide an executive dashboard with AI Action Center, Demand card, and Subsystem Health strip.

5. Technology Stack

Layer	Technology	Purpose in PharmaShift
Frontend	React 18.2 + Vite	Single Page Application, reactive components
Styling / UI	Tailwind CSS + Recharts	Responsive dark theme, charts & SVG topology
Backend	FastAPI 0.110 (Python 3.11+)	Asynchronous REST API, OpenAPI docs
Database	SQLite / PostgreSQL 15	Relational persistence with SQLAlchemy ORM
Machine Learning	scikit-learn (Random Forest)	Demand forecasting regression (MAE: 1.2152, R²: 0.7523)
Security	PyJWT + bcrypt	Stateless JWT auth, salted password hashing
Testing	pytest + httpx	30 automated test cases (100% pass rate)

6. Project Milestones & Development Progress

The development progress is tracked across 10 functional phases:

Milestone Phase	Major Functional Scope	Status
Phase 1: Foundation	Mathematical excess model, 5,193 synthetic batches	COMPLETED
Phase 2: Database & Backend	10 SQLAlchemy ORM models, FastAPI core, auto-seeding	COMPLETED
Phase 3: Inventory Network	18 branches, Haversine distance & transit days math	COMPLETED
Phase 4: Recommender Engine	Excess calculation with 3-day safety buffer, feasibility filters	COMPLETED
Phase 5: ML Demand Model	Random Forest Regressor trained (MAE: 1.2152, R²: 0.7523)	COMPLETED
Phase 6: ML + Recommender	Live ML inference in destination ranking with O(1) cache	COMPLETED (Review #2)
Phase 7: Frontend Dashboard	AI Action Center hero card, Demand card, Priority Queue table	COMPLETED (Review #2)
Phase 8: HITL & Audit Trail	Approval checklist, structured rejections, overrides, audit logs	COMPLETED (Review #2)
Phase 9: Testing & Edge Cases	30 automated tests (100% pass), 8-scenario Edge Cases Sandbox	COMPLETED (Review #2)
Phase 10: Analytics & Benchmark	30-scenario simulation suite (+█7.44L gain, -52.09% loss)	COMPLETED (Review #2)

7. Review #1 vs Review #2 Improvements

Detailed comparison of improvements implemented after Review #1:

Area	Review #1 State	Review #2 State	Improvement
ML Integration	Model trained but static heuristics used in ranking	Live RF predictions in candidate ranking with cache	Active predictive ML scoring
Explainability	Generic text bullet list	Structured Safety & Feasibility checklist bullets	Auditable clinical checklist
Scoring	Uncalibrated 'confidence %'	Normalized Recommendation Score (0-100)	Eliminated probability claim
Seed Security	POST /api/seed unprotected	Guarded with require_roles(['ADMIN'])	401/403/200 RBAC enforced
Reference Date	Volatile today() runtime	DEMO_REFERENCE_DATE env (default 2026-08-14)	Deterministic evaluation
Telemetry	Static {'status': 'ok'}	SystemHealthOut with ML metrics, DB, Ref Date	Full subsystem visibility
Dashboard UI	Static KPI cards only	AI Action Center, Demand card, Priority table	Immediate clinical actionability
Test Suite	21 test cases	30 test cases (100% passing in 76.87s)	+9 tests covering ML & RBAC
Documentation	Minor number variations	Synchronized with simulation_results.json	Single source of truth

8. Machine Learning Demand Forecasting

A **RandomForestRegressor** (100 trees, max depth 10) was trained on 1,080 historical demand records across 18 branches and 60 medicines. The model ingests 11 engineered features: weekly average daily dispense (42.1% importance), monthly average (28.4%), storage capacity (14.2%), unit price (9.1%), baseline rate (3.8%), volatility (1.2%), and days-to-expiry (0.6%).

Metric	Value	Source Artifact	Interpretation
Mean Absolute Error (MAE)	1.2152 units/day	ml/model_metrics.json	Average daily error ~1.2 units
Root Mean Squared Error (RMSE)	3.5170 units/day	ml/model_metrics.json	Low variance in prediction errors
R ² Goodness of Fit	0.7523	ml/model_metrics.json	75.23% demand variance explained
In-Memory Cache Latency	< 0.05 ms	ml_service._prediction_cache	O(1) memory lookup for destinations

9. Empirical Simulation Results (Baseline vs Proposed)

Across 30 multi-scenario simulation cycles (1,794 total recommendations evaluated), the proposed system achieved significant improvements over the naive isolated FIFO baseline:

Evaluation Metric	Baseline (Isolated FIFO)	Proposed Recommender	Net Gain / Impact
Average Value Protected	■75,70,344.23	■83,14,345.38	+■7,44,001.15 (+9.88%)
Value Lost to Expiry	■14,28,450.00	■6,84,448.85	-■7,44,001.15 (-52.09% loss reduction)
Acceptance Rate	N/A	93.03%	Simulation-based acceptance behavior
Average Transit Distance	N/A	10.27 km	Urban logistics efficiency (Haversine)
Average Shelf Life at Transfer	N/A	19.41 days	Ample buffer before expiration

10. Automated Testing Summary (30 Tests Passed)

Executed `pytest backend/tests/test\_backend.py -v`: **30 passed, 0 failed in 76.87s (100% pass rate)**. Test coverage encompasses expiry math, risk classification, safety buffers, Haversine routing, feasibility blocks, JWT authentication, human approval, mandatory rejection reasons, quantity overrides, audit logging, RBAC restrictions, ML inference, ML fallback, reference dates, seed security (401/403/200), evidence checklists, and health telemetry.

11. Remaining Work for Review #3 (Final 30%)

The remaining ~21.5% of development required for full production readiness includes:

- 1. Hospital & Pharmacy ERP Connectors (HL7 / FHIR live database ingestion).
- 2. Capacitated Vehicle Routing Problem (CVRP) multi-stop delivery solver.
- 3. Automated MLOps continuous retraining pipeline with model drift alerts.
- 4. Multi-tenant database isolation and SAML 2.0 / OAuth2 corporate Single Sign-On.
- 5. Mobile pharmacist barcode/RFID scanner application for warehouse receipt.

12. Conclusion

PharmaShift has successfully advanced from its Review #1 baseline to an integrated, explainable, and machine-learning-driven pharmacy stock redistribution platform. With **78.5% completion**, 30 passing automated tests, a zero-error production build, and empirical validation proving a +9.88% value gain and 52.09% expiry waste reduction, the project firmly meets and exceeds all criteria for Review #2.